Matplotlib

matplotlib is a python library it is used for data visualisation

importing matplotlib

In [1]:
```
import matplotlib.pyplot as plt
import numpy as np
```

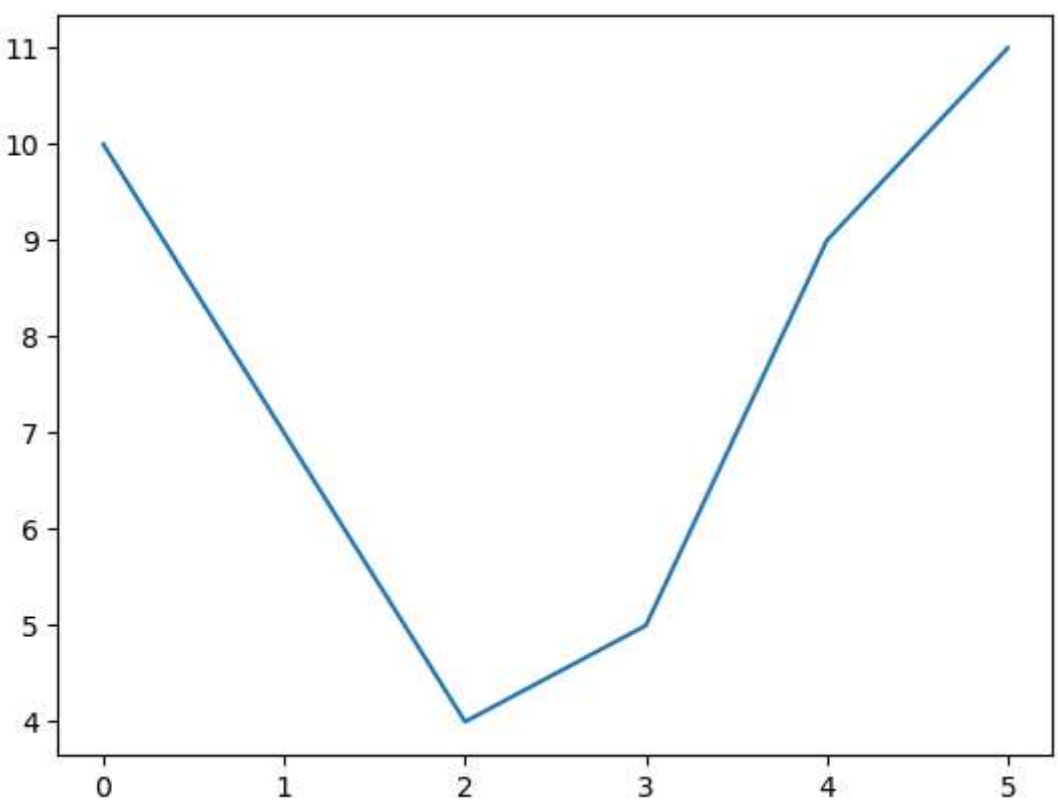
Essential Role of Pyplot in Matplotlib

pyplot is a module in matplotlib it provide functions like plot(),bar(),pie(),hist(),scatter() functions helps us to make plots easily

Line Graph

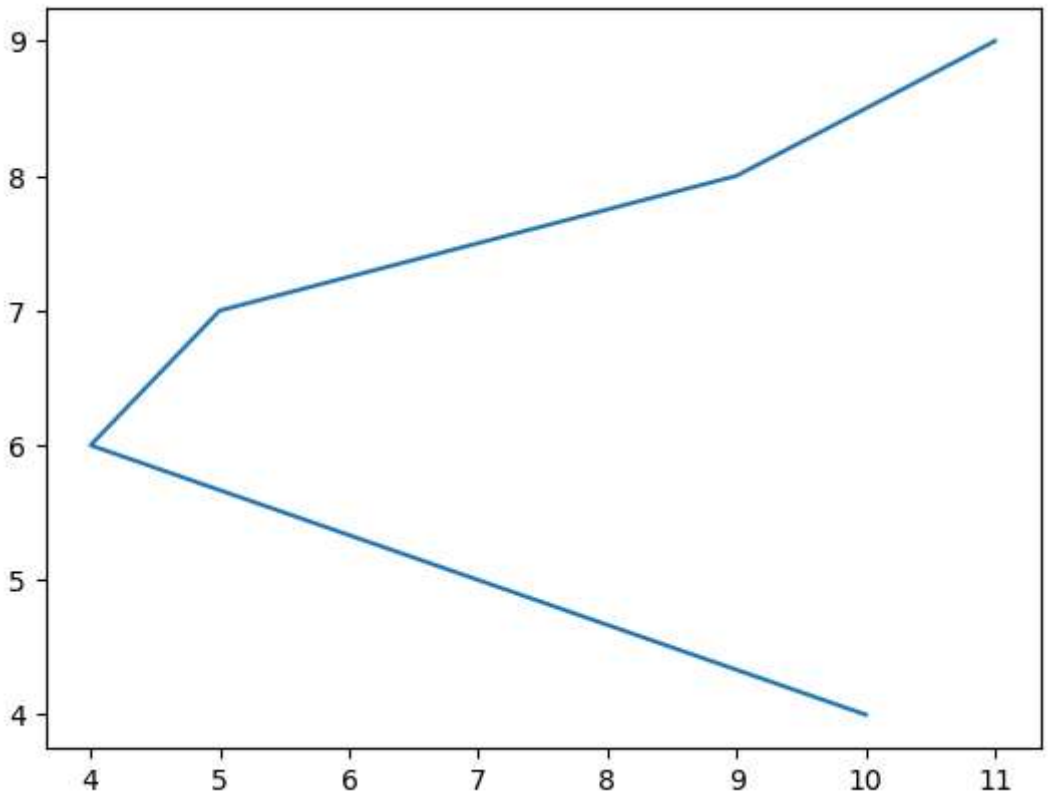
Line graph is used when you want to show continuous data and how it is changing over trends

In [2]:
```
x=np.array([10,7,4,5,9,11])
plt.plot(x)
plt.show()
```



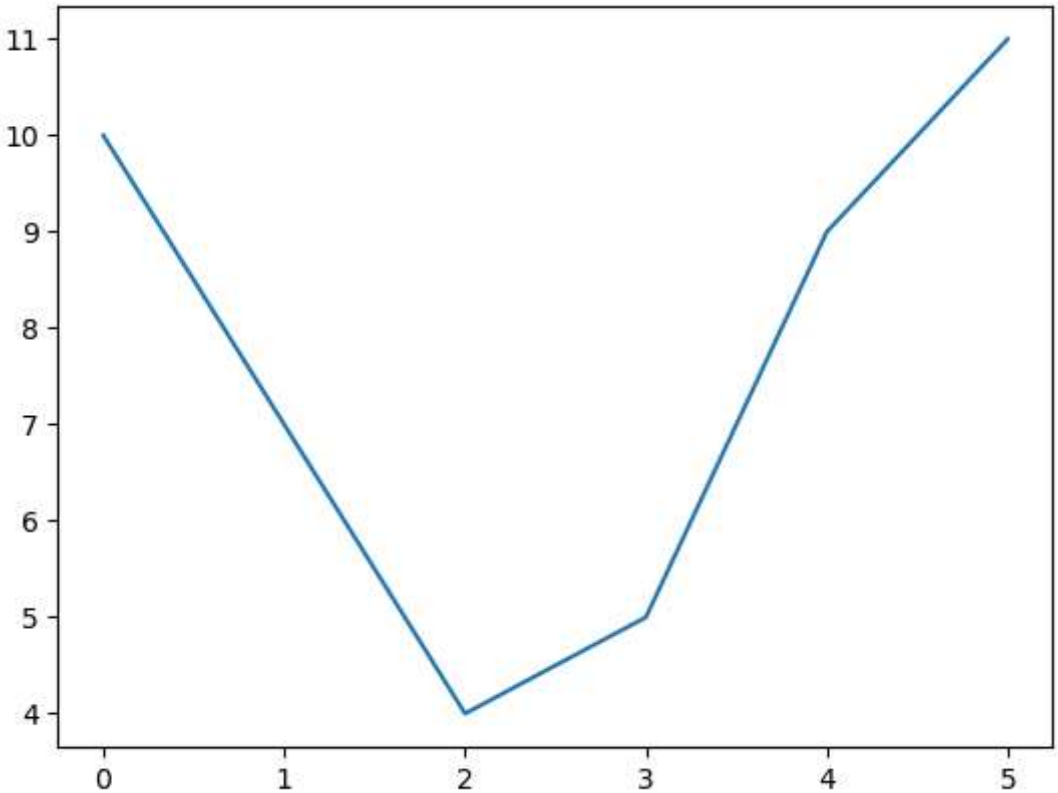
only given y-axis variables we plot the and consider x axis as index of y-axis variable

In [3]:
```
x=np.array([10,7,4,5,9,11])
y=np.array([4,5,6,7,8,9])
plt.plot(x,y)
plt.show()
```

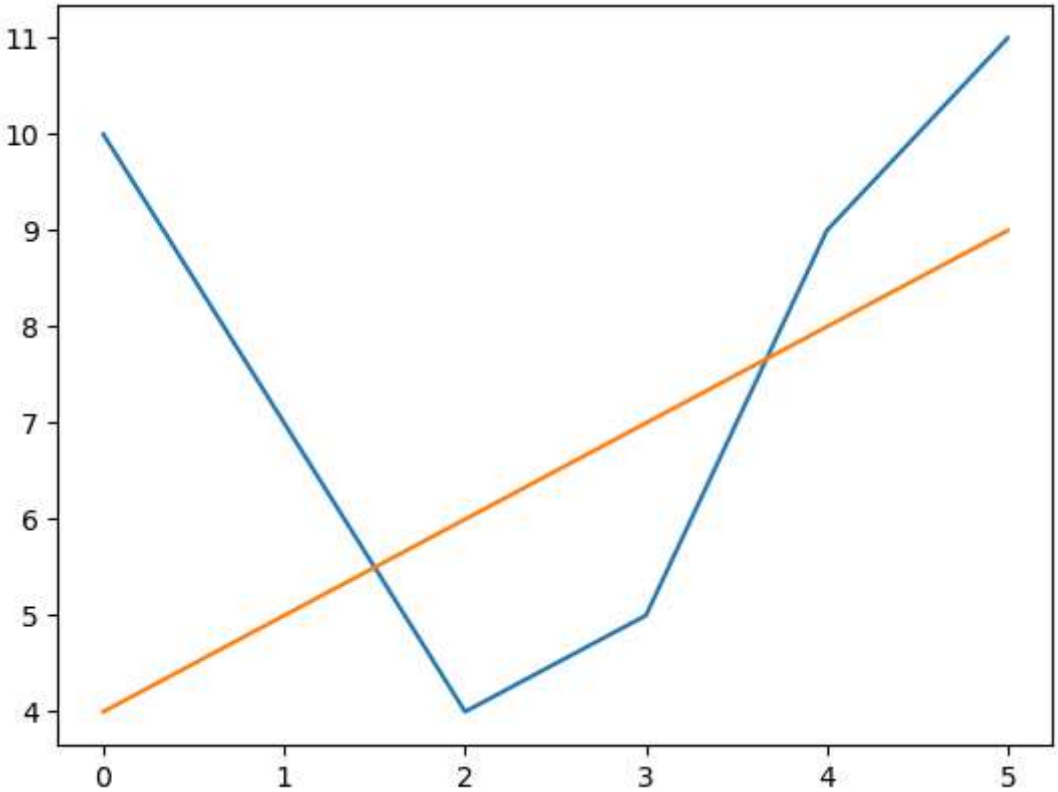


In [4]:
```
x1=np.array([10,7,4,5,9,11])
x2=np.array([4,5,6,7,8,9])
plt.plot(x1)
```

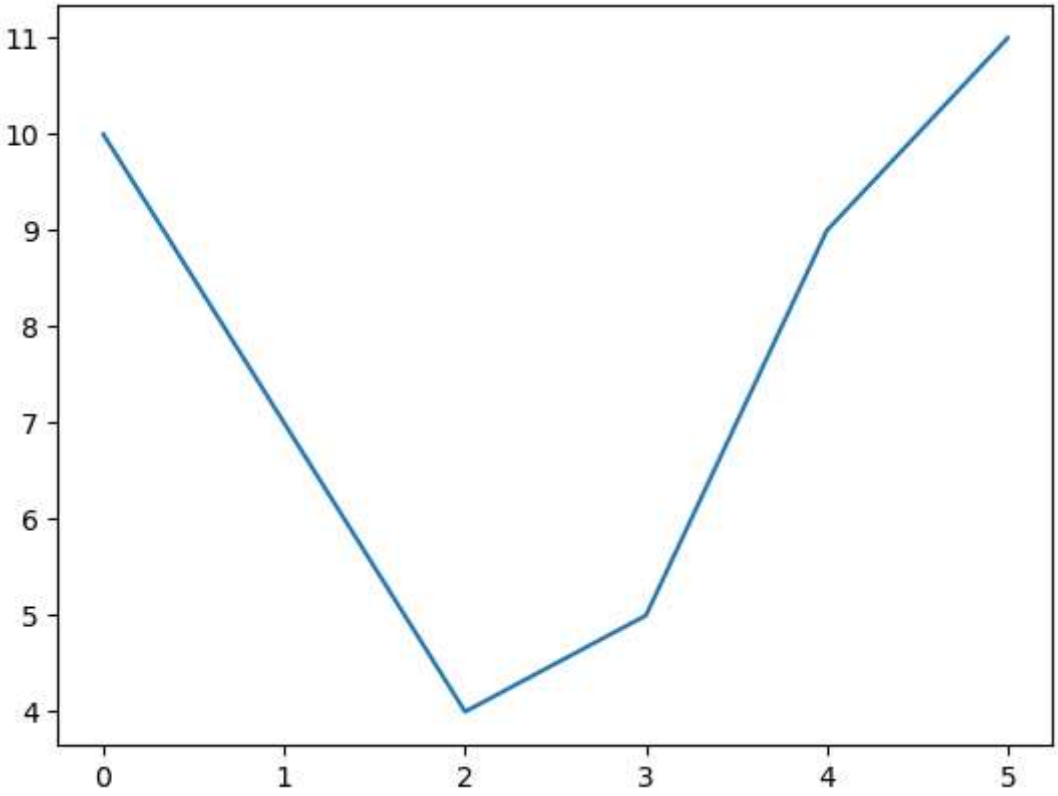
Out[4]:
```
[<matplotlib.lines.Line2D at 0x13ca458f610>]
```

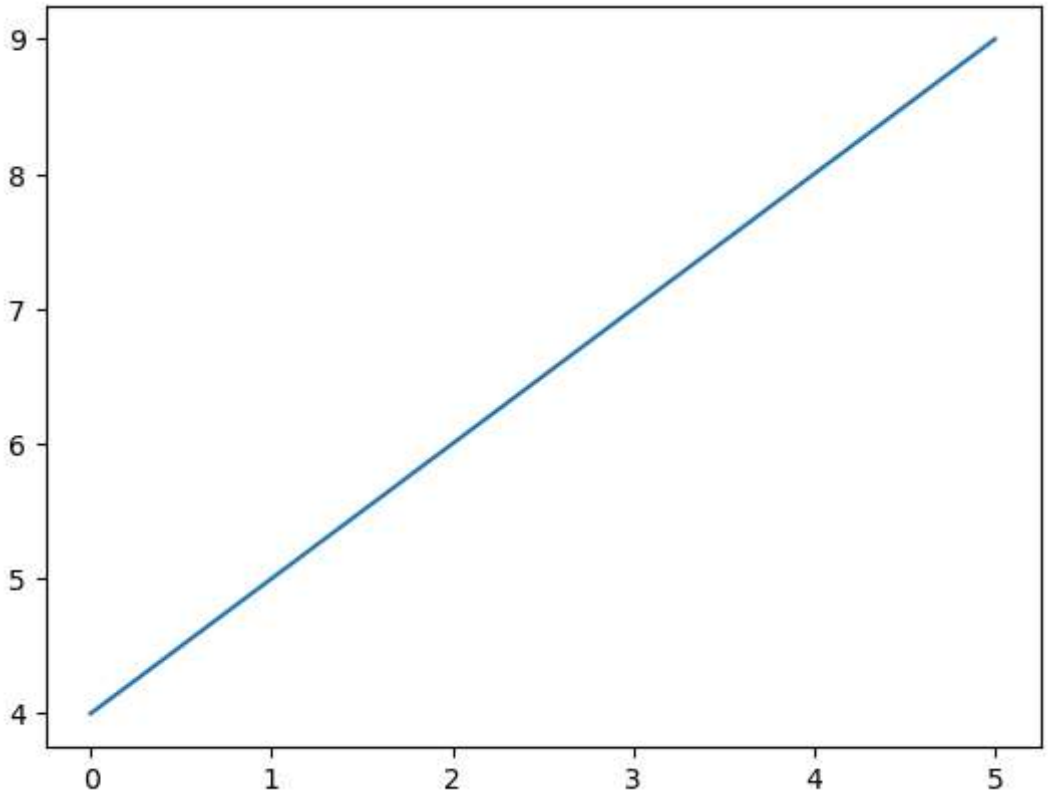


In [5]:
```
x1=np.array([10,7,4,5,9,11])
x2=np.array([4,5,6,7,8,9])
plt.plot(x1)
plt.plot(x2)
plt.show()
```

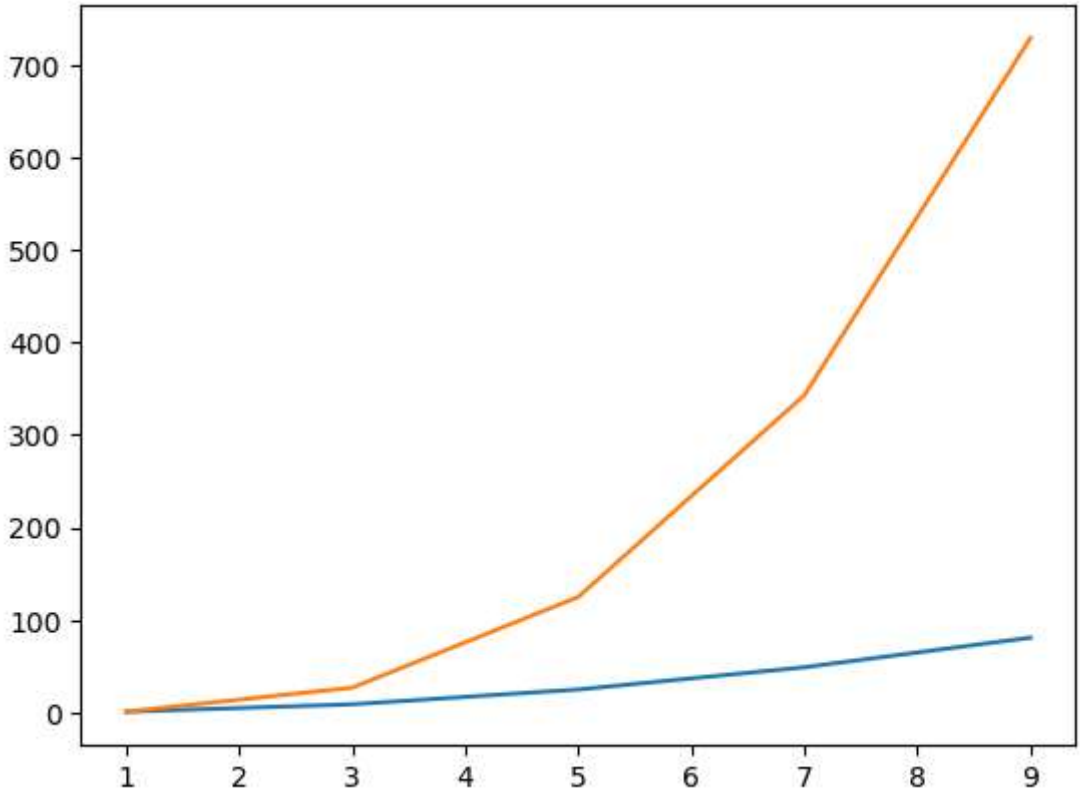


In [6]:
```
x1=np.array([10,7,4,5,9,11])
plt.plot(x1)
plt.show()
x2=np.array([4,5,6,7,8,9])
plt.plot(x2)
plt.show()
```



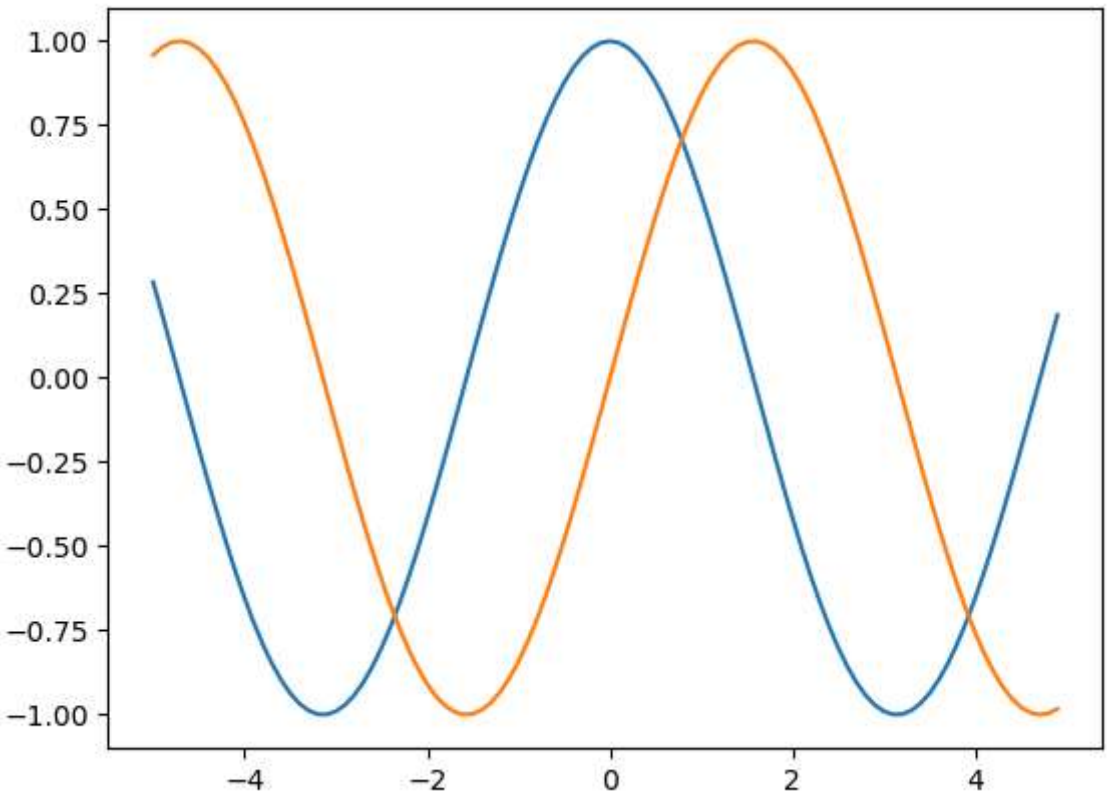


In [7]:
```
x=np.array([1,3,5,7,9])
plt.plot(x,x**2)
plt.plot(x,x**3)
plt.show()
```



in plt.plot pass arguments based on our requirement if we pass one argument it consider by default on y axis it plot x axis based on indexes

In [8]:
```
x=np.arange(-5,5,0.1)
plt.plot(x,np.cos(x))
plt.plot(x,np.sin(x))
plt.show()
```

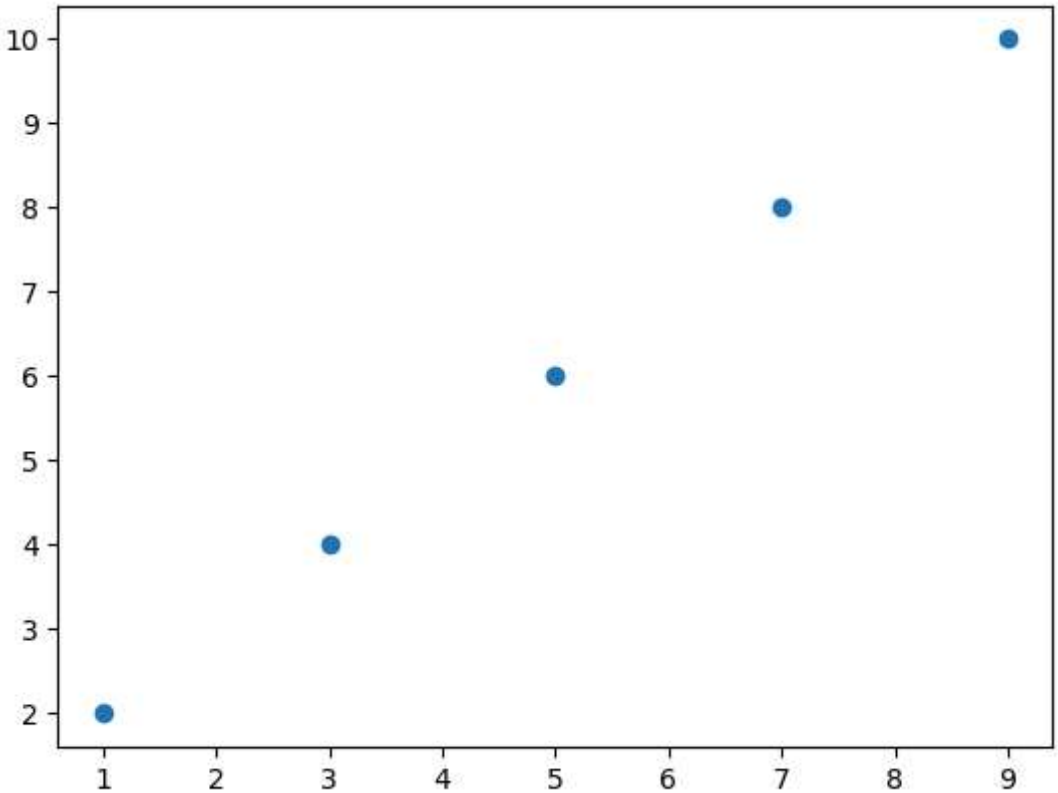


scatter plot

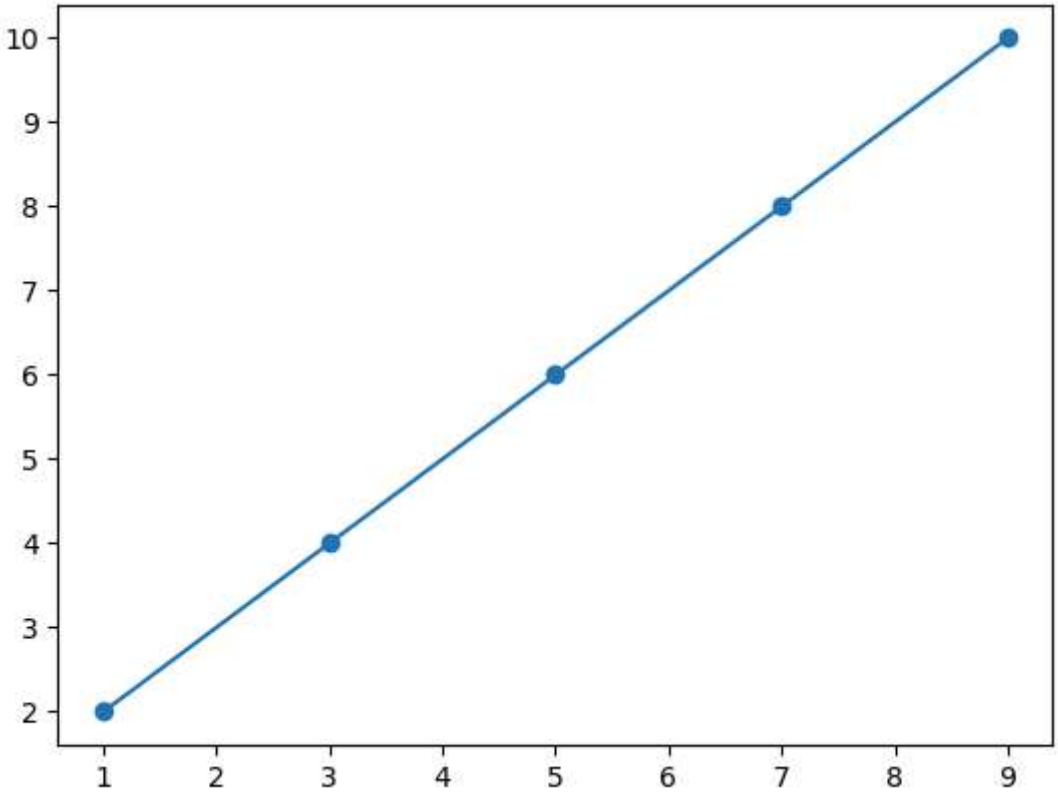
scatter plot is used for visualizing relation between two continuous variables and it is used to find outliers, correlation

in scatter plot we have to give both x and y arguments

In [9]:
```
x=np.array([1,3,5,7,9])
y=np.array([2,4,6,8,10])
plt.scatter(x,y)
plt.show()
```

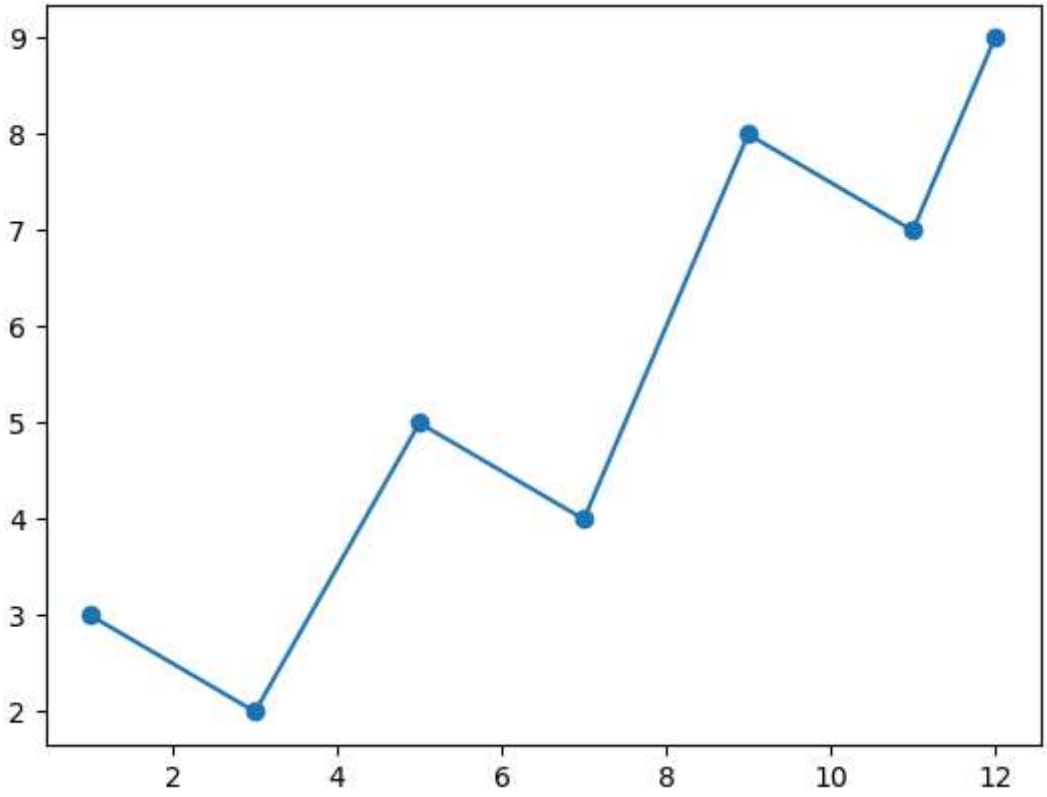


In [10]:
```
x=np.array([1,3,5,7,9])
y=np.array([2,4,6,8,10])
plt.scatter(x,y)
plt.plot(x,y)
plt.show()
```

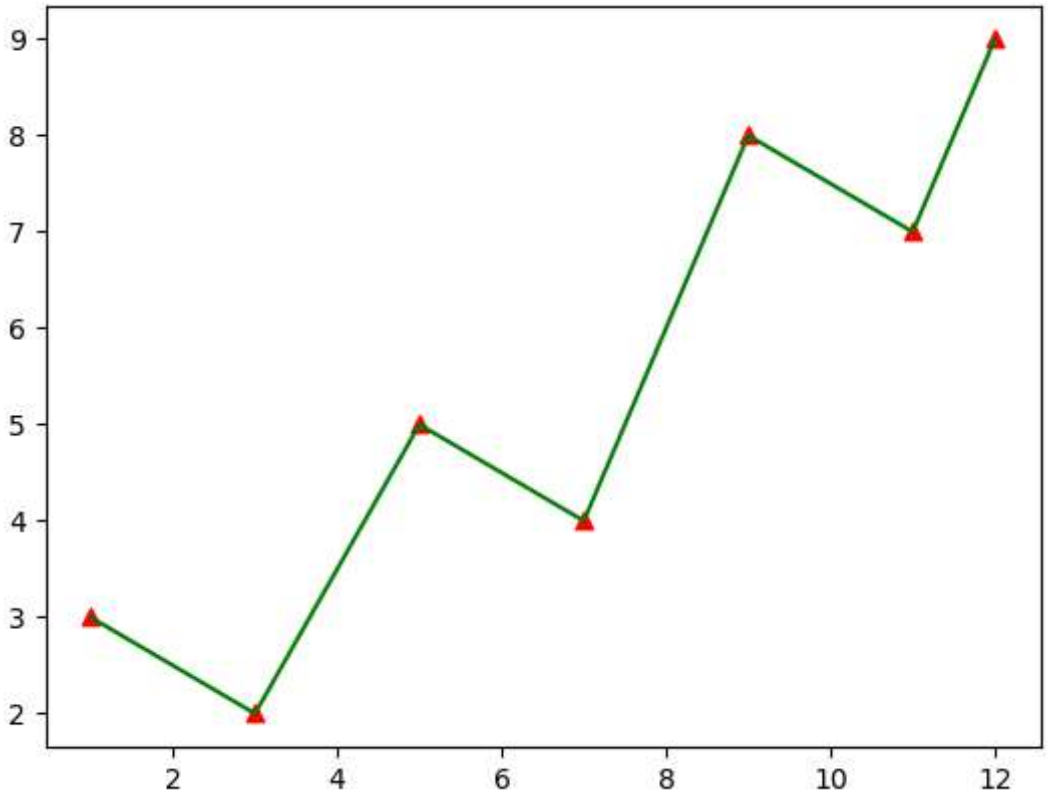


we can use both scatter and plt.plot function to get this kind of graph

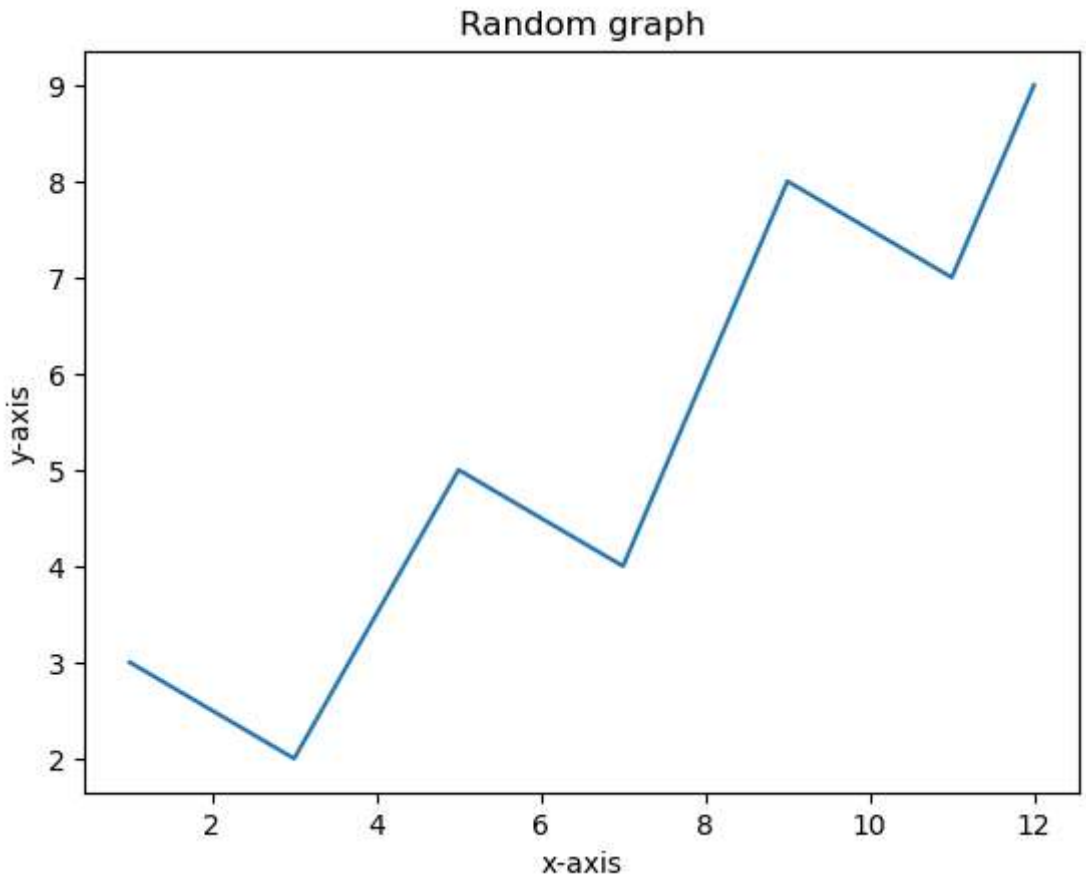
In [11]:
```
x=np.array([1,3,5,7,9,11,12])
y=np.array([3,2,5,4,8,7,9])
plt.scatter(x,y)
plt.plot(x,y)
plt.show()
```



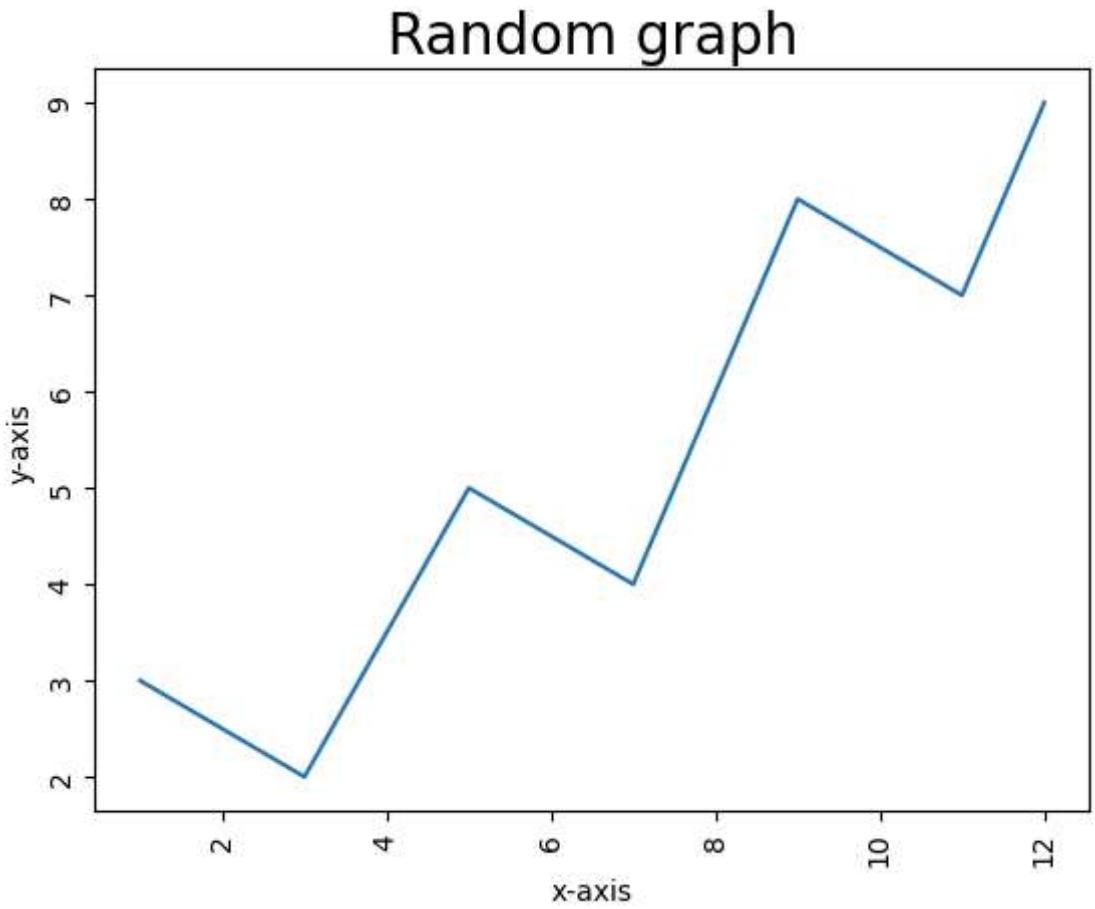
In [12]:
```
x=np.array([1,3,5,7,9,11,12])
y=np.array([3,2,5,4,8,7,9])
plt.scatter(x,y,color='red',marker='^')
plt.plot(x,y,color='green')
plt.show()
```



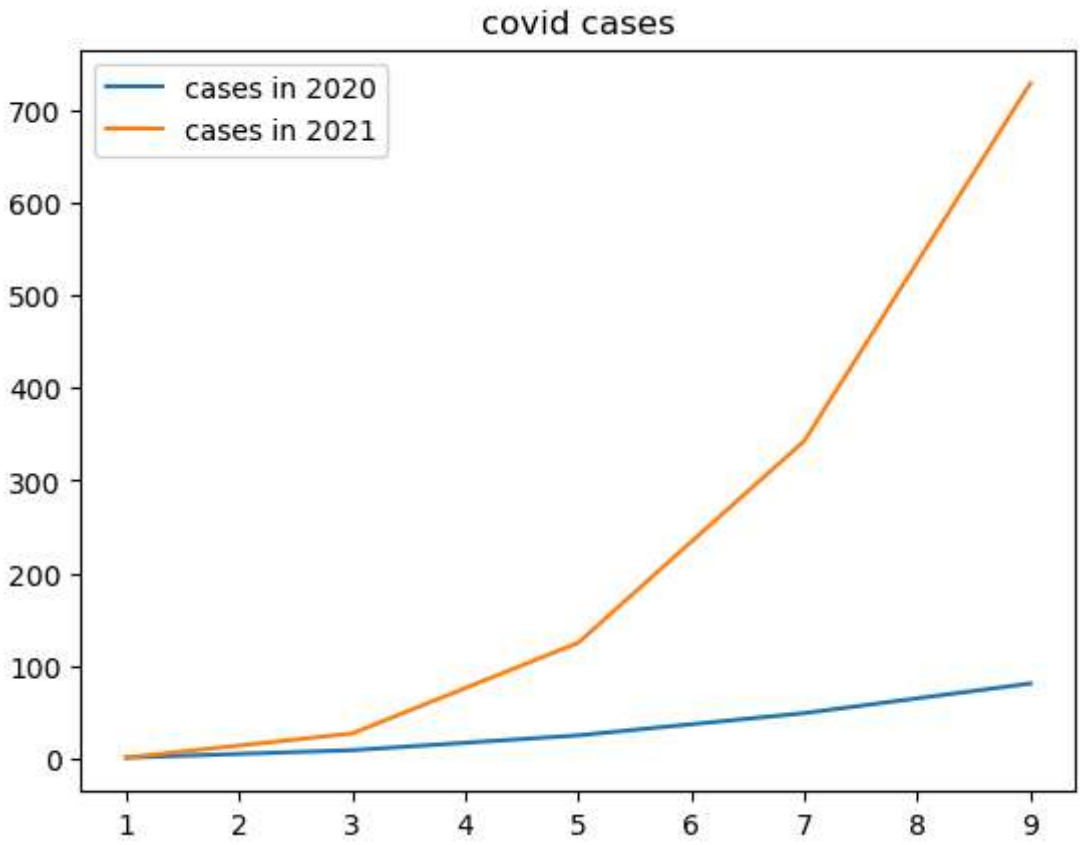
In [13]:
```
x=np.array([1,3,5,7,9,11,12])
y=np.array([3,2,5,4,8,7,9])
plt.title("Random graph")
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.plot(x,y)
plt.show()
```



In [14]:
```
x=np.array([1,3,5,7,9,11,12])
y=np.array([3,2,5,4,8,7,9])
plt.title("Random graph",fontsize=20)
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.xticks(rotation=90)
plt.yticks(rotation=90)
plt.plot(x,y)
plt.show()
```

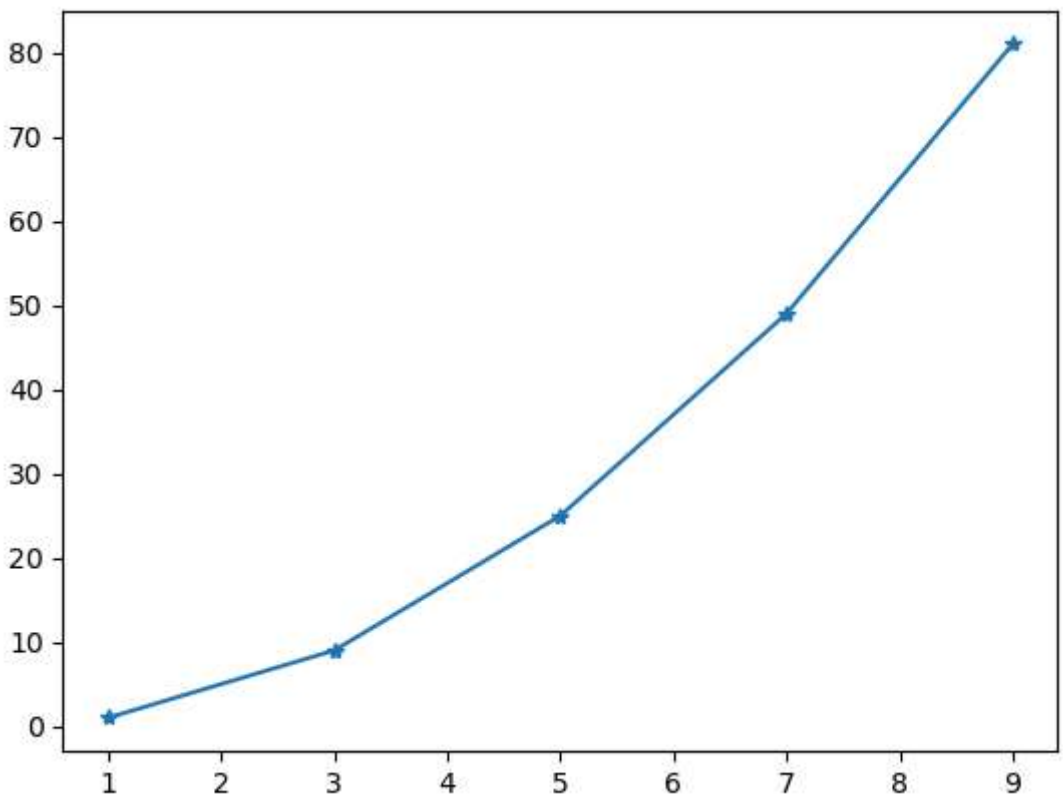


In [15]:
```
x=np.array([1,3,5,7,9])
plt.plot(x,x**2,label="cases in 2020")
plt.plot(x,x**3,label="cases in 2021")
plt.title("covid cases")
plt.legend()
plt.show()
```



In [16]:
```
x=np.array([1,3,5,7,9])
plt.plot(x,x**2,label="cases in 2020",marker="*")

plt.show()
```

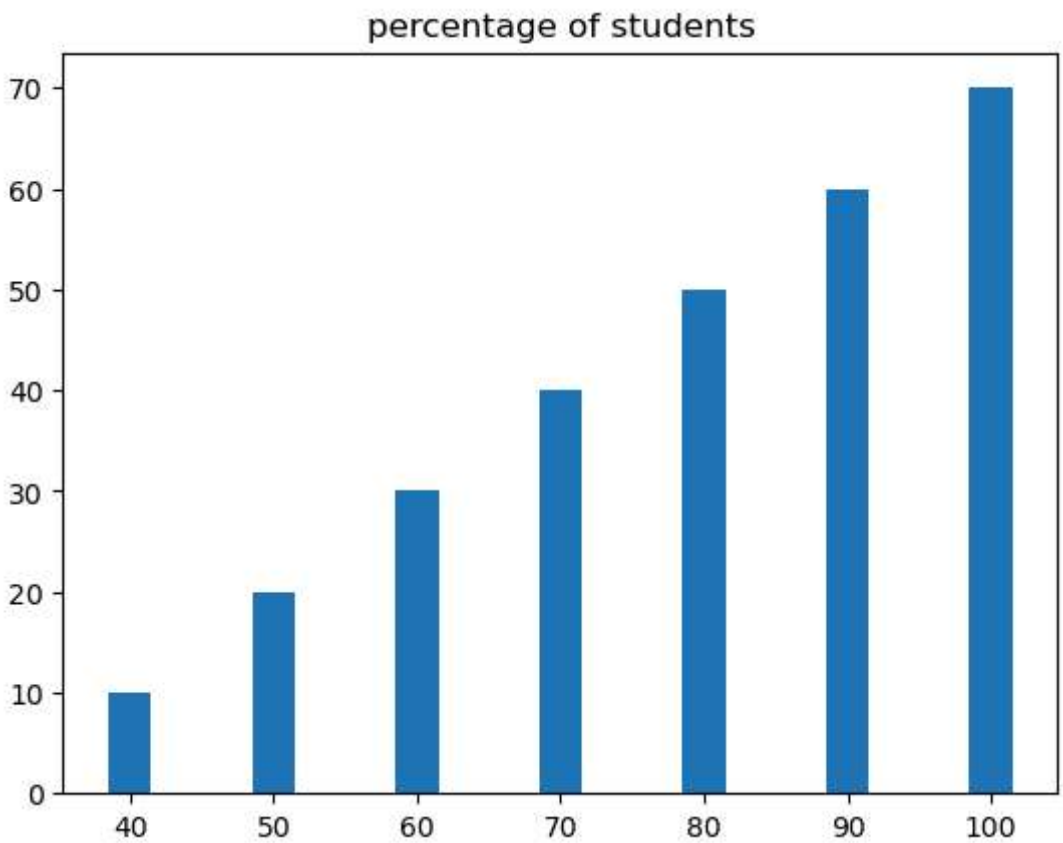


bar plot

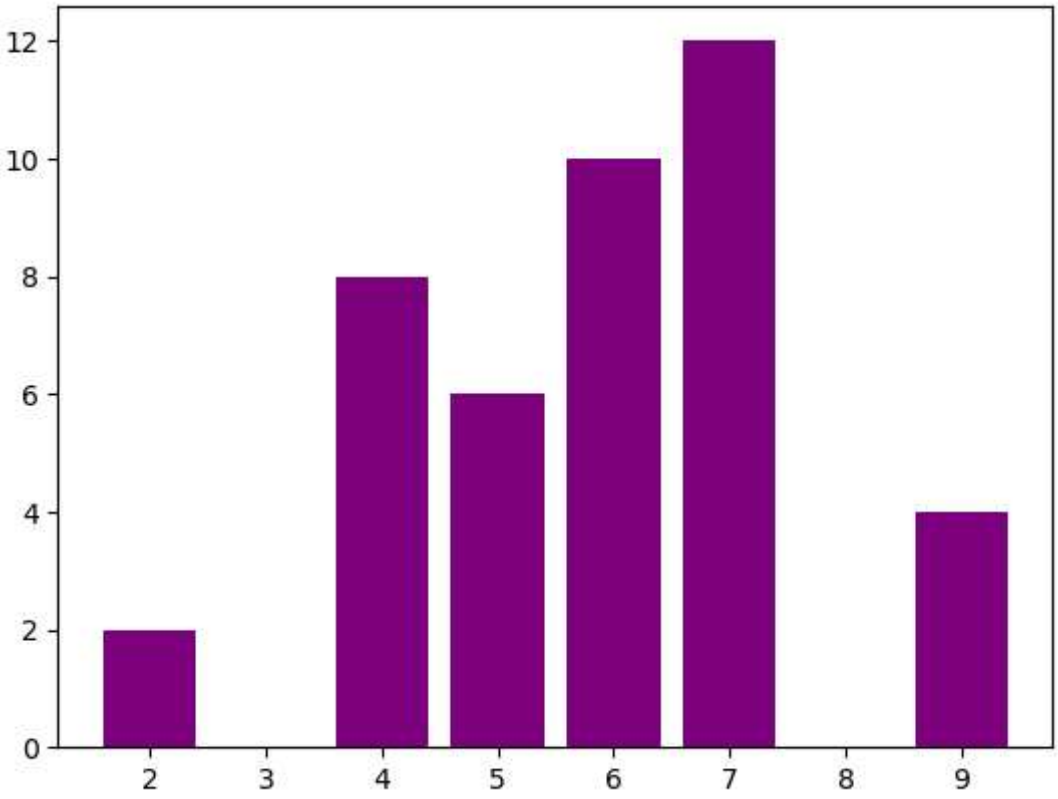
bar chart is mainy used for categorical data and for comparing differnt categories

bar chart is a graph that uses rectangular bars to represent data

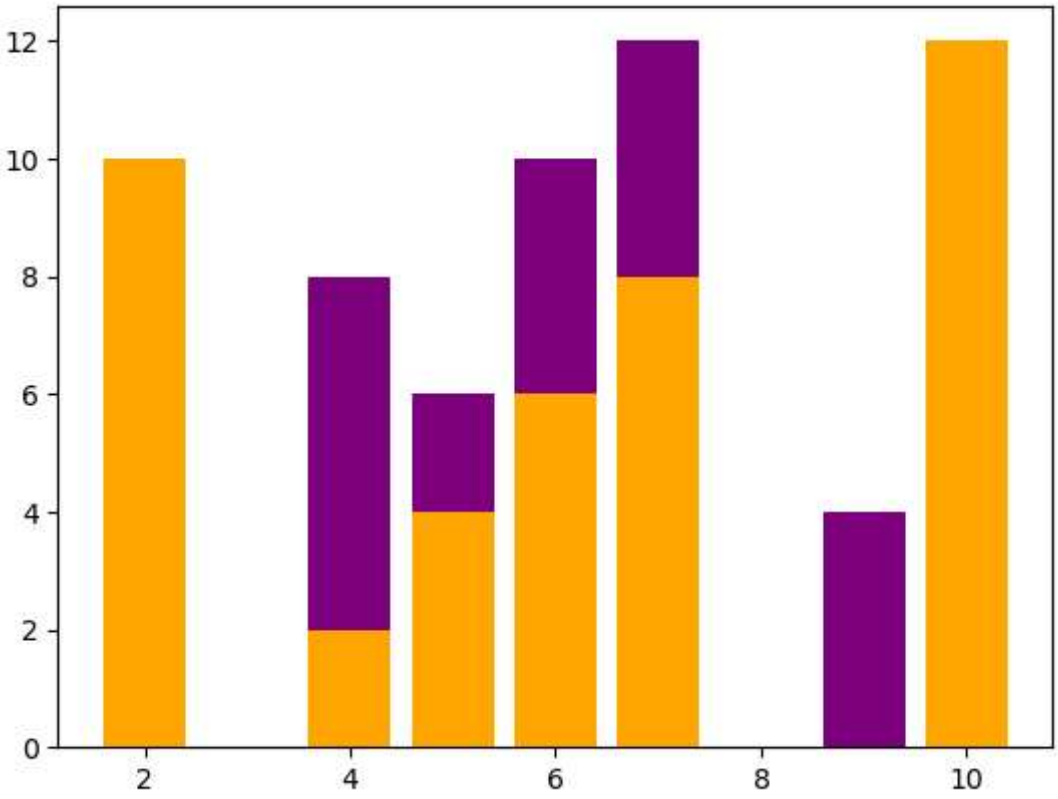
In [17]:
```
percent_marks=np.array([40,50,60,70,80,90,100])
num_students=np.array([10,20,30,40,50,60,70])
plt.title("percentage of students")
plt.bar(percent_marks,num_students,width=3)
plt.show()
```



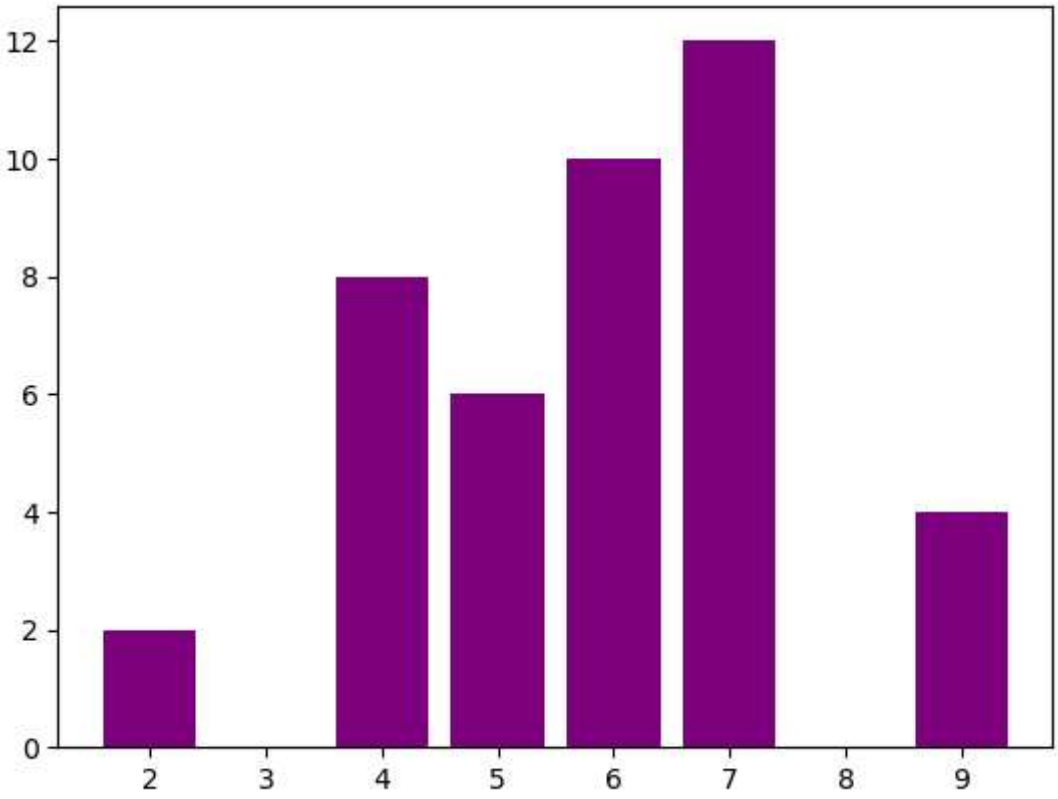
In [18]:
```
x1=np.array([2,9,5,4,6,7])
y1=np.array([2,4,6,8,10,12])
plt.bar(x1,y1,color="purple")
plt.show()
```

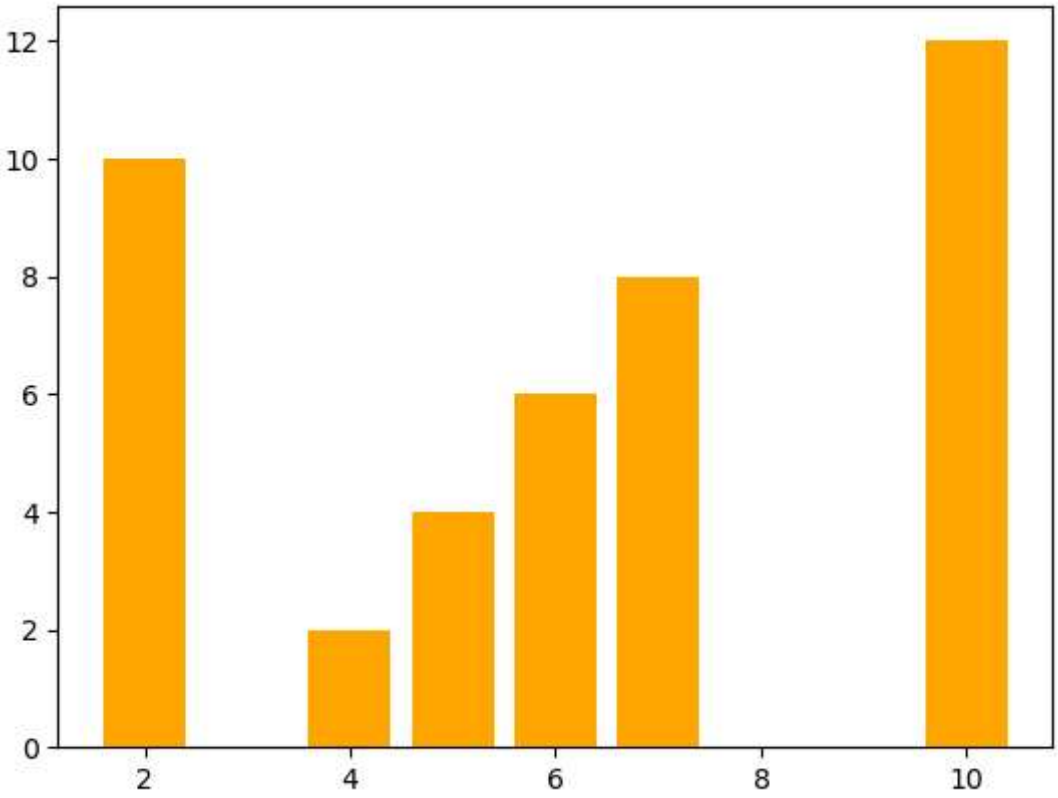


In [19]:
```
x1=np.array([2,9,5,4,6,7])
y1=np.array([2,4,6,8,10,12])
plt.bar(x1,y1,color="purple")
x2=np.array([4,5,6,7,2,10])
plt.bar(x2,y1,color='orange')
plt.show()
```

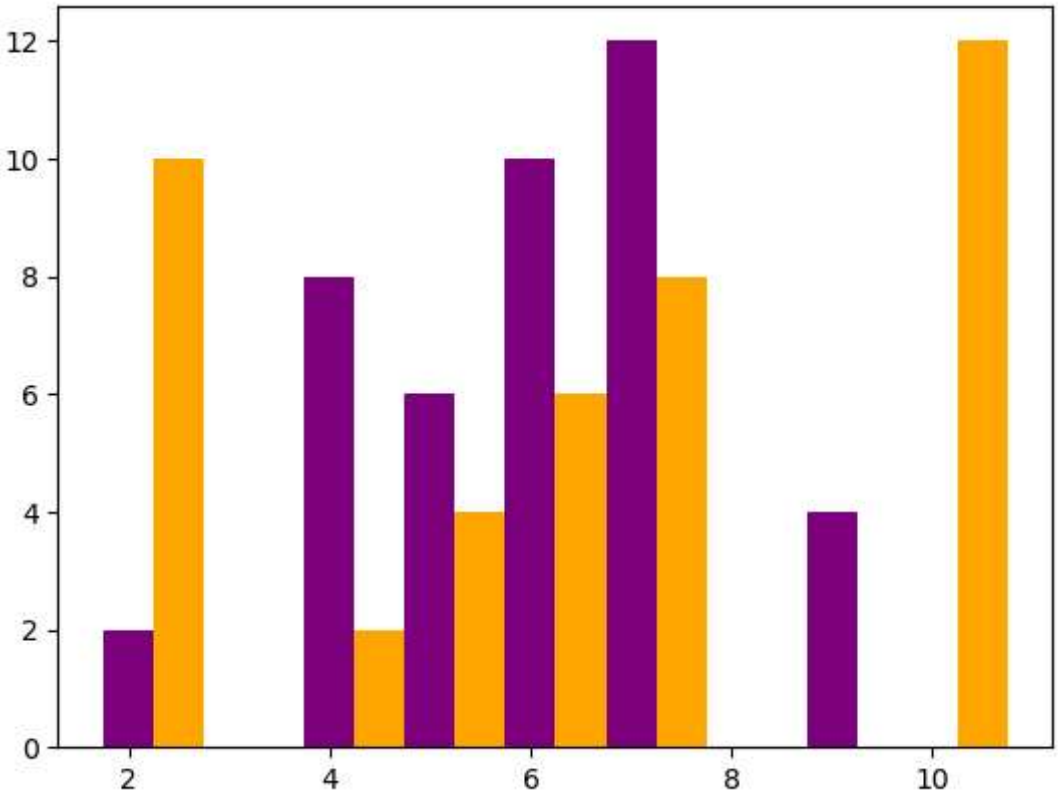


In [20]:
```
x1=np.array([2,9,5,4,6,7])
y1=np.array([2,4,6,8,10,12])
plt.bar(x1,y1,color="purple")
plt.show()
x2=np.array([4,5,6,7,2,10])
plt.bar(x2,y1,color='orange')
plt.show()
```



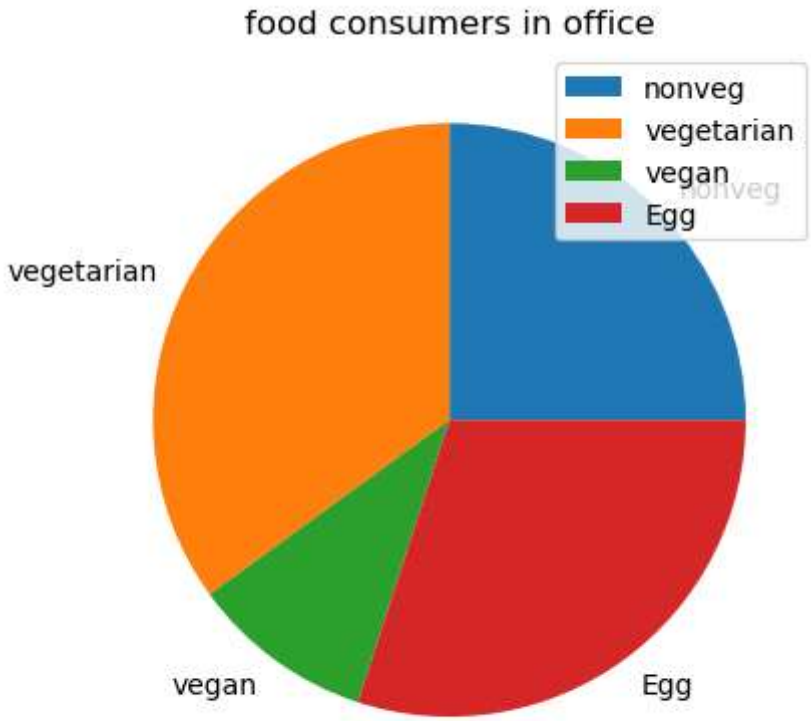


In [21]:
```
x1=np.array([2,9,5,4,6,7])
y1=np.array([2,4,6,8,10,12])
plt.bar(x1,y1,color="purple",width=0.5)
x2=np.array([4,5,6,7,2,10])
plt.bar(x2+0.5,y1,width=0.5,color='orange')
plt.show()
```

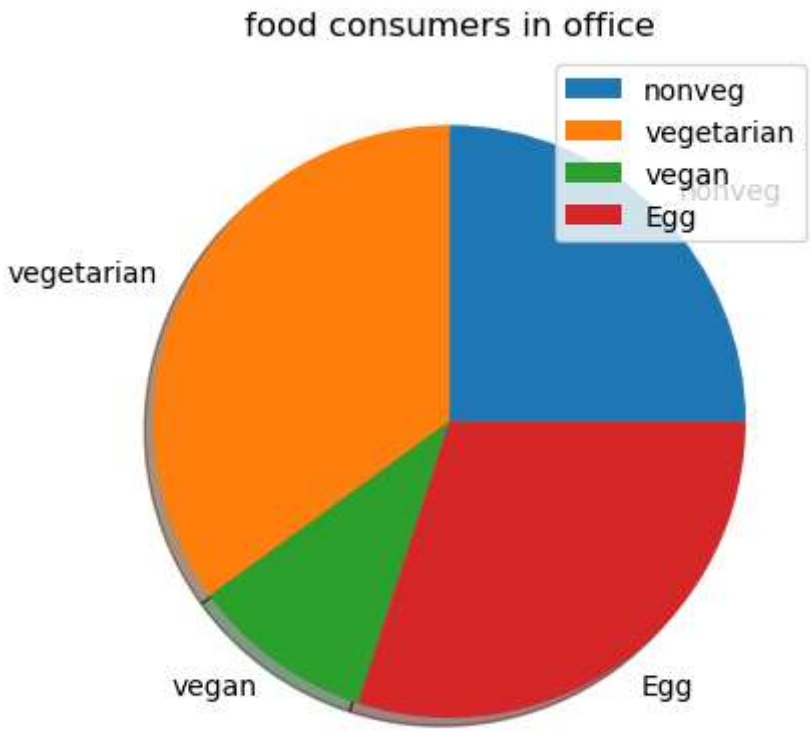


pie graph

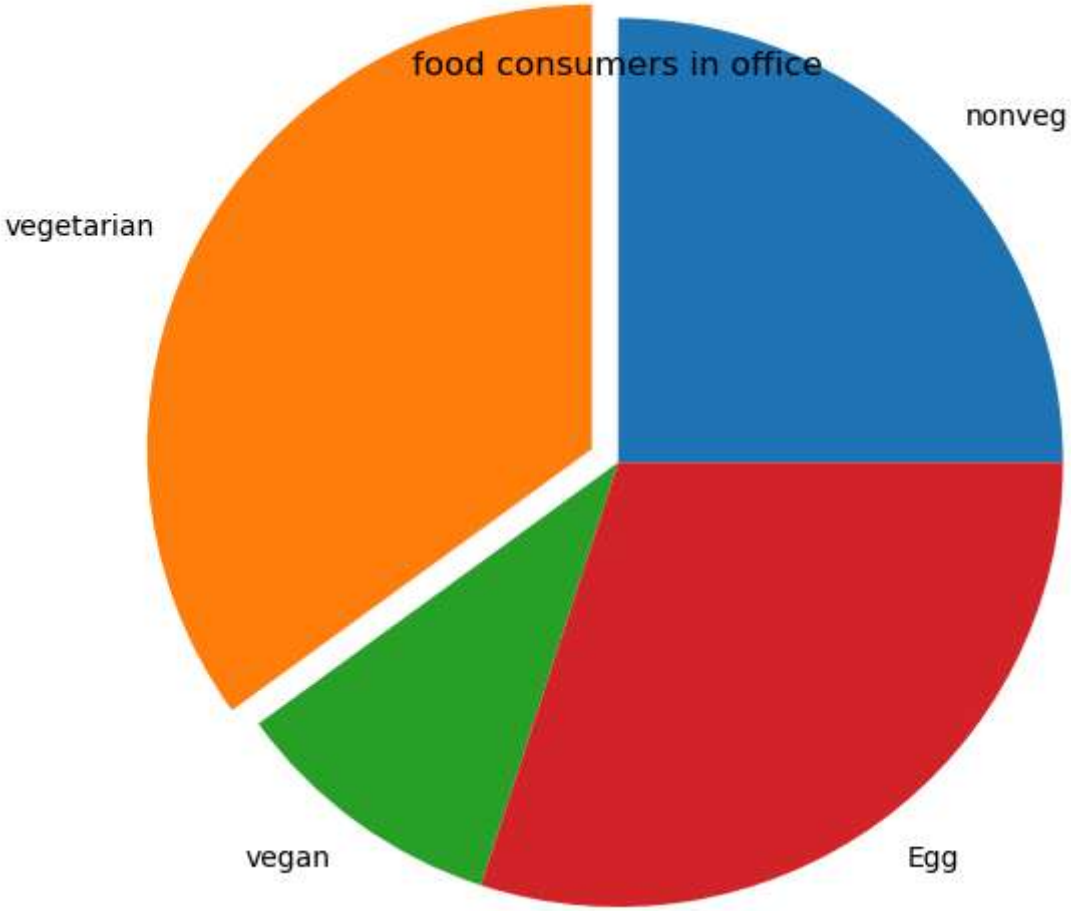
In [22]:
```
values=[25,35,10,30]
label=["nonveg","vegetarian","vegan","Egg"]
plt.pie(values,labels=label)
plt.legend()
plt.title("food consumers in office")
plt.show()
```



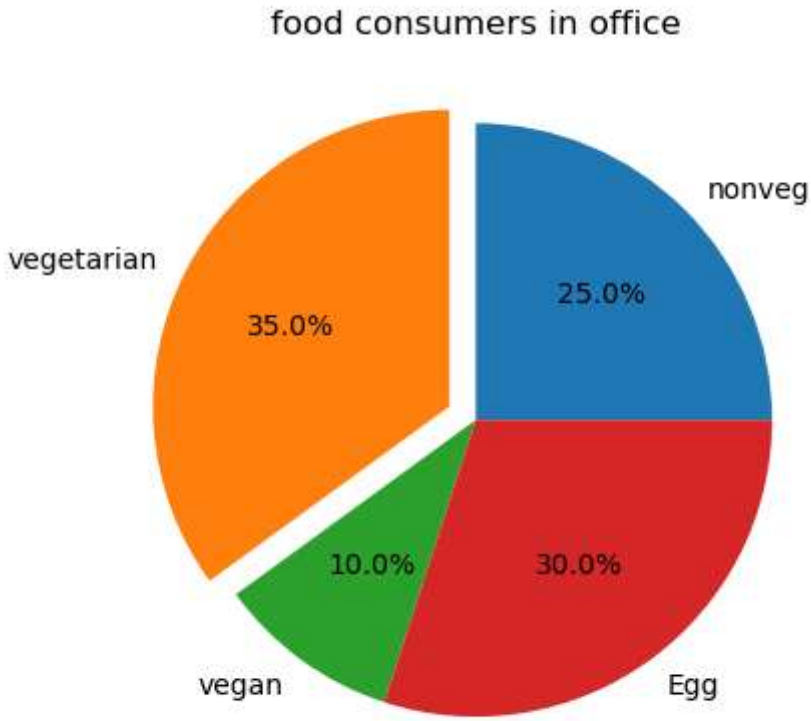
In [23]:
```
values=[25,35,10,30]
label=["nonveg","vegetarian","vegan","Egg"]
plt.pie(values,labels=label,shadow=True)
plt.legend()
plt.title("food consumers in office")
plt.show()
```



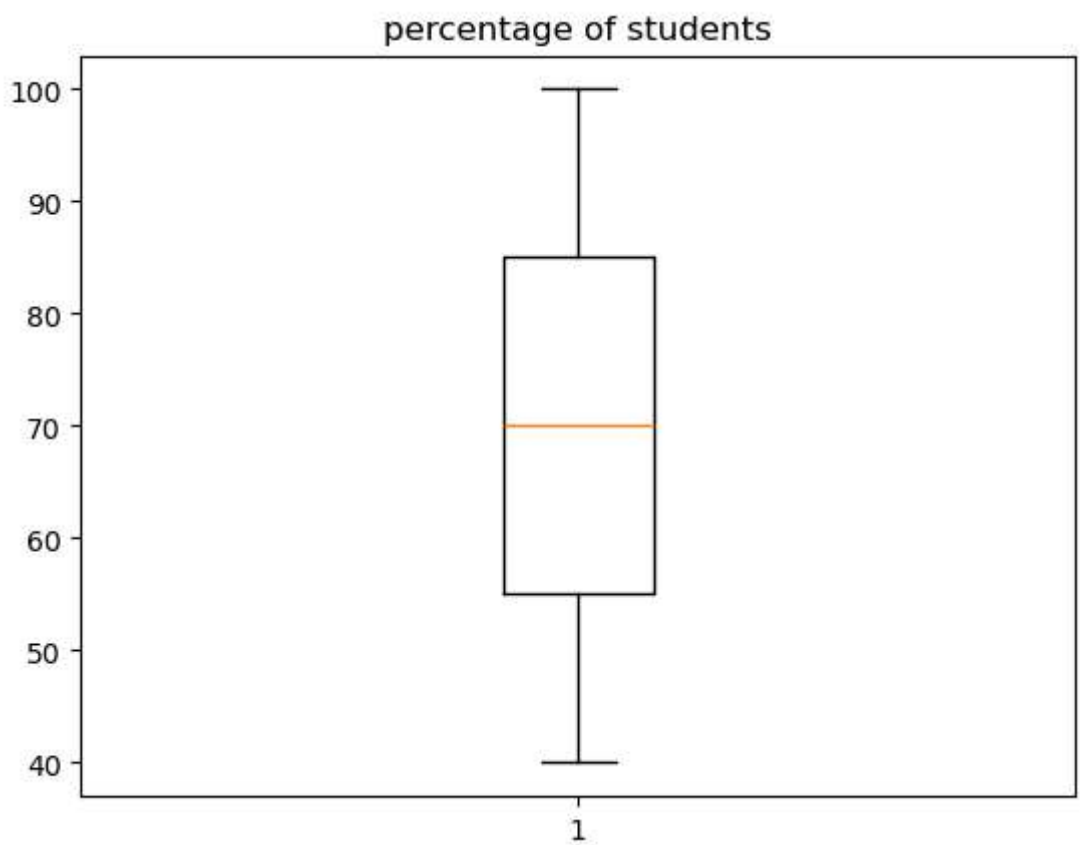
In [24]:
```
values=[25,35,10,30]
label=["nonveg","vegetarian","vegan","Egg"]
plt.pie(values,labels=label,explode=(0,0.1,0,0),radius=1.5)
plt.title("food consumers in office")
plt.show()
```



In [25]:
```
values=[25,35,10,30]
label=["nonveg","vegetarian","vegan","Egg"]
plt.pie(values,labels=label,explode=(0,0.1,0,0),radius=1,autopct='%1.1f%%')
plt.title("food consumers in office")
plt.show()
```



In [28]:
```
percent_marks=np.array([40,50,60,70,80,90,100])
plt.title("percentage of students")
plt.boxplot(percent_marks)
plt.show()
```



minimum-smallest value-40

first quartile(Q1)-25 percentile-55

median(Q2)-50th percentile-70

third quartile(Q3)-85

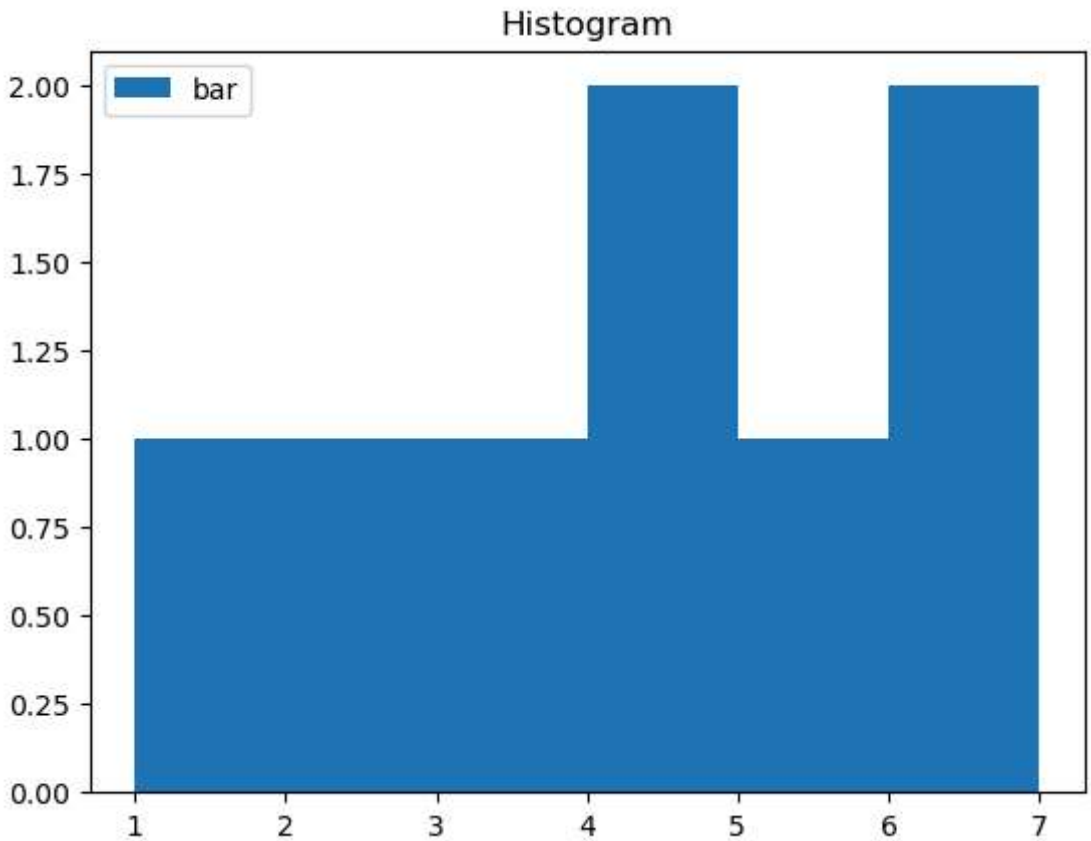
maximun-(Q4)-100

In [1]:
```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5, 6, 7, 4]

plt.hist(x, bins = [1, 2, 3, 4, 5, 6, 7])

plt.title("Histogram")
plt.legend(["bar"])

plt.show()
```



In [ ]: